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System administration assignment two

Introduction

In this assignment, I worked with Linux, web servers, Docker, and VirtualBox. I installed and set up a web server and database on a remote server, created Docker images, and tested a Linux system on a virtual machine. I used tools like Google to search for help and guides, and the website and documentation for Ubuntu, Apache, MariaDB, Docker, and VLC.

Requirements 1

First I moved the .pem key into the ~/.ssh/ directory and restricted using chmod 400, and then connected successfully to the server using SSH.

```
osama@OSAMAH:~$ mv ~/oa222sv-key-pair.pem ~/.ssh/  
osama@OSAMAH:~$ chmod 400 ~/.ssh/oa222sv-key-pair.pem  
osama@OSAMAH:~$ ssh -i ~/.ssh/oa222sv-key-pair.pem ubuntu@194.47.177.247
```

To ensure the server was secure and updated, I ran update commands using apt. This ensures packages are at their latest versions.

```
ubuntu@oa222sv-server:~$ sudo apt update  
Get:1 http://education.clouds.archive.ubuntu.com/ubuntu noble InRelease [256  
kB]  
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease  
ubuntu@oa222sv-server:~$ sudo apt upgrade -y  
Reading package lists... Done  
Building dependency tree... Done
```

In this step I checked information about the server. First, I checked the system version, and I found it's Ubuntu 24.04.2 with Linux kernel 6.8.

```
ubuntu@oa222sv-server:~$ lsb_release -a  
No LSB modules are available.  
Distributor ID: Ubuntu  
Description:    Ubuntu 24.04.2 LTS  
Release:        24.04  
Codename:       noble  
ubuntu@oa222sv-server:~$ uname -a  
Linux oa222sv-server 6.8.0-62-generic #65-Ubuntu SMP PREEMPT_DYNAMIC Mon May  
19 17:15:03 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux  
ubuntu@oa222sv-server:~$
```

Then I checked the disk usage using df -h, and I saw how much space was free and used. I also looked at open ports with ss -tuln, to see what services are running. Finally, I listed the users on the server with a command to read the /etc/passwd file.

```

ubuntu@oa222sv-server:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        8.7G  867M   7.8G  10% /
tmpfs            985M   0    985M   0% /dev/shm
tmpfs            394M  680K   393M   1% /run
tmpfs            5.0M   0    5.0M   0% /run/lock
/dev/sda16       881M  133M   687M  17% /boot
/dev/sda15       105M   6.2M   99M    6% /boot/efi
tmpfs            197M   8.0K   197M   1% /run/user/1000
ubuntu@oa222sv-server:~$ sudo ss -tuln
Netid  State      Recv-Q  Send-Q      Local Address:Port  Peer Address:Port
udp    UNCONN     0        0          127.0.0.54:53       0.0.0.0:*
udp    UNCONN     0        0          127.0.0.53%lo:53    0.0.0.0:*
udp    UNCONN     0        0      192.168.0.35%ens3:68 0.0.0.0:*
tcp    LISTEN     0      4096          127.0.0.54:53       0.0.0.0:*
tcp    LISTEN     0      4096      127.0.0.53%lo:53    0.0.0.0:*
tcp    LISTEN     0      4096             *:22                 *:22
ubuntu@oa222sv-server:~$ cut -d: -f1 /etc/passwd
root
daemon
bin
sys
sysrc

```

I installed MariaDB as the database server for saving and reading data. and Apache as the web server, this allows the server to show websites when users visit my IP. After that, I set Apache to start automatically and confirmed it was running.

```

ubuntu@oa222sv-server:~$ sudo apt install mariadb-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:

```

```

ubuntu@oa222sv-server:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:

```

```

ubuntu@oa222sv-server:~$ sudo systemctl start apache2
ubuntu@oa222sv-server:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@oa222sv-server:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-06-28 16:32:21 UTC; 43s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 7965 (apache2)
      Tasks: 55 (limit: 2335)
     Memory: 5.2M (peak: 5.5M)
        CPU: 40ms
    CGroup: /system.slice/apache2.service

```

I added a text 'Hello World' to check everything worked.

```

ubuntu@oa222sv-server:~$ echo "Hello World" | sudo tee /var/www/html/index.html
Hello World

```



The screenshot shows a web browser window with the address bar displaying '194.47.177.247'. The page content is 'Hello World'.

Then I enabled CGI in Apache to run scripts in the browser.

```
ubuntu@oa222sv-server:~$ sudo a2enmod cgi
Your MPM seems to be threaded. Selecting cgid instead of cgi.
Enabling module cgid.
To activate the new configuration, you need to run:
    systemctl restart apache2
ubuntu@oa222sv-server:~$ sudo a2ensite default
ERROR: Site default does not exist!
ubuntu@oa222sv-server:~$ sudo systemctl restart apache2
```

I created a Bash script called info.bash that shows system details like uptime and disk space.

```
GNU nano 7.2 /usr/lib/cgi-bin/info.bash *
#!/bin/bash
echo "Content-type: text/plain"
echo ""
echo "Server Info:"
echo "-----"
uname -a
df -h
uptime
who
free -h
```

I wrote a backup script that saves my website and Apache settings in a compressed file.

```
GNU nano 7.2 /home/ubuntu/backup.bash
#!/bin/bash
tar -czvf backup.tar.gz /var/www /etc/apache2
```

```
ubuntu@oa222sv-server:~$ nano ~/backup.bash
ubuntu@oa222sv-server:~$ chmod +x ~/backup.bash
ubuntu@oa222sv-server:~$ ./backup.bash
tar: Removing leading '/' from member names
/var/www/
/var/www/html/
/var/www/html/index.html
tar: Removing leading '/' from hard link targets
/etc/apache2/
```

I tested the info.bash script in the browser, this proves CGI setup worked and the script gave data through the browser.

```
← → ↻ ⚠ Not secure 194.47.177.247/cgi-bin/info.bash ☆ 📁 👤 ⋮
```

```
Server Info:
-----
Linux oa222sv-server 6.8.0-62-generic #65-Ubuntu SMP PREEMPT_DYNAMIC Mon May 19 17:15:03 UTC 2025 x86_64
x86_64 x86_64 GNU/Linux
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        8.7G  1.3G  7.4G  15% /
tmpfs            985M   0  985M   0% /dev/shm
tmpfs            394M 704K  393M   1% /run
tmpfs            5.0M   0   5.0M   0% /run/lock
tmpfs            197M  8.0K  197M   1% /run/user/1000
/dev/sda16       881M 133M  687M  17% /boot
/dev/sda15       105M  6.2M   99M   6% /boot/efi
 18:35:52 up 7 days, 11:43,  3 users,  load average: 0.00, 0.00, 0.00
ubuntu pts/0    Jun 28 18:31 (192.168.240.11)
ubuntu pts/2    Jun 28 16:30 (192.168.240.11)
ubuntu pts/3    Jun 28 16:51 (192.168.240.11)
ubuntu pts/4    Jun 28 16:54 (192.168.240.11)
ubuntu pts/5    Jun 28 16:55 (192.168.240.11)
ubuntu pts/6    Jun 28 17:28 (192.168.240.11)

```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	341Mi	918Mi	1.0Mi	870Mi	1.6Gi
Swap:	0B	0B	0B			

Requirements 2

First, I installed docker and enabled it and started on the server. And I created a new directory called cowsay-image and entered it, this is where I would build the Docker image for the cowsay program.

```
ubuntu@oa222sv-server:~$ sudo systemctl enable docker
ubuntu@oa222sv-server:~$ sudo systemctl start docker
ubuntu@oa222sv-server:~$ mkdir cowsay-image
ubuntu@oa222sv-server:~$ cd cowsay-image
```

Inside cowsay-image, I created a Dockerfile

```
ubuntu@oa222sv-server:~/cowsay-image$ nano Dockerfile
ubuntu@oa222sv-server:~/cowsay-image$ nano docker-compose.yaml
```

and wrote command tells Docker to use Debian and install cowsay tool inside the container.

```
GNU nano 7.2 Dockerfile
FROM debian:bullseye

RUN apt update && apt install -y cowsay

ENTRYPOINT ["cowsay"]
```

I made another file called docker-compose.yaml, This helps to run the container. In this file, I tell Docker to use version 3 of the format. I implemented a service called cow.

Then I give it a command to run when it starts, it uses Bash to run cowsay, and prints 'Hello from Docker!' in a fun way.

```
GNU nano 7.2 docker-compose.yaml
version: '3'
services:
  cow:
    build: .
    entrypoint: ["bash", "-c", "/usr/games/cowsay Hello from Docker!"]
```

With both files ready, I ran:

```
ubuntu@oa222sv-server:~/cowsay-image$ docker-compose up
WARN[0000] /home/ubuntu/cowsay-image/docker-compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 1/1
✔ Container cowsay-image-cow-1 Rec... 0.2
Attaching to cow-1
cow-1 | -----
cow-1 | < Hello from Docker! >
cow-1 | -----
cow-1 |          ^__^
cow-1 |         (oo)\_____
cow-1 |        (__)\       )\/\
cow-1 |           ||----w |
cow-1 |           ||
cow-1 | exited with code 0
ubuntu@oa222sv-server:~/cowsay-image$
w Enable Watch
```

Next, I created a second Docker image that serves a CGI script using Apache.

I created a new directory, Then I created a Dockerfile again:

```
ubuntu@oa222sv-server:~$ mkdir apache-cgi
ubuntu@oa222sv-server:~$ cd apache-cgi
ubuntu@oa222sv-server:~/apache-cgi$ mkdir cgi-bin
ubuntu@oa222sv-server:~/apache-cgi$ nano cgi-bin/info.bash
ubuntu@oa222sv-server:~/apache-cgi$ chmod +x cgi-bin/info.bash
ubuntu@oa222sv-server:~/apache-cgi$ nano Dockerfile
```

```

GNU nano 7.2 info.bash
#!/bin/bash
echo "Content-type: text/plain"
echo ""
echo "Server Info:"
echo "-----"
uname -a
df -h
uptime
who
free -h

```

This installs Apache and enables CGI, and copies my script info.bash to the CGI folder.

```

GNU nano 7.2 Dockerfile *
FROM ubuntu:24.04

RUN apt update && apt install -y apache2

RUN apt install -y apache2-bin apache2-utils apache2-suexec-pristine \
    apache2-data libapache2-mod-perl2 libapache2-mod-php \
    libapache2-mod-fcgid

RUN apt install -y bash curl

# Enable CGI
RUN a2enmod cgi

# Create and copy script into cgi-bin
COPY cgi-bin/info.bash /usr/lib/cgi-bin/info.bash
RUN chmod +x /usr/lib/cgi-bin/info.bash

# Expose port
EXPOSE 80

# Start Apache in foreground
CMD ["apachectl", "-D", "FOREGROUND"]

```

I created a separate docker-compose.yaml, This maps container port 80 to host port 80, allowing web access to the container.

```

GNU nano 7.2 docker-compose.yaml
services:
  apache-cgi:
    build: .
    ports:
      - "80:80"

```

After running docker-compose up --build again, I accessed the CGI script by visiting <http://localhost/cgi-bin/info.bash>, the script executed and returned all the expected system info, confirming that CGI was working within the container.

```

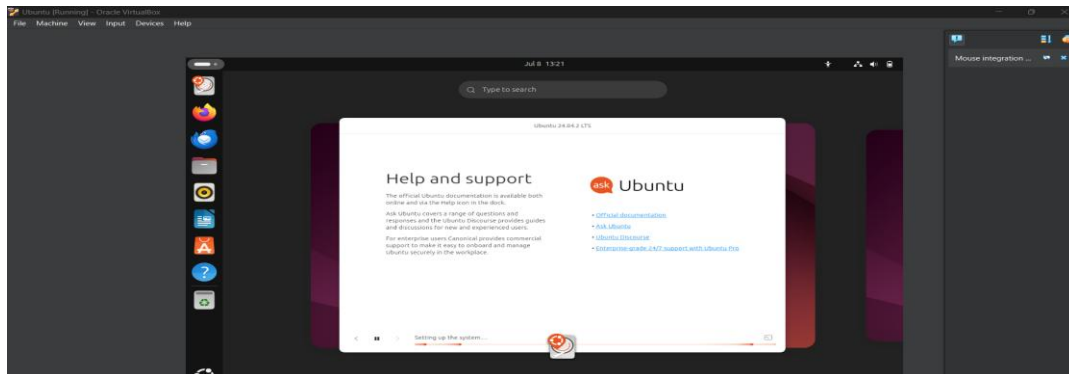
ubuntu@oa222sv-server:~/apache-cgi$ docker exec -it apache-cgi-apache-cgi-1
bash
root@75a8e243014f:/# curl http://localhost/cgi-bin/info.bash
Server Info:
-----
Linux 75a8e243014f 6.8.0-62-generic #65-Ubuntu SMP PREEMPT_DYNAMIC Mon May 19 17:15:03 UTC 2025 x86_64 x86_64 GNU/Linux
Filesystem      Size      Used Avail Use% Mounted on
overlay          8.7G      2.5G      6.2G 29% /
tmpfs            64M        0        64M   0% /dev
shm             1.2M        0        1.2M   0% /dev/shm
/dev/root        8.7G      2.5G      6.2G 29% /etc/hosts
tmpfs            985M        0      985M   0% /proc/acpi
tmpfs            985M        0      985M   0% /proc/scsi
tmpfs            985M        0      985M   0% /sys/firmware
14:45:50 up 9 days, 7:53,  0 user, load average: 0.00, 0.00, 0.00
             total    used    free   shared  buff/cache   avail
Mem:          1.9Gi       501Mi       309Mi       4.7Mi       1.3Gi       1
Swap:          0B           0B           0B
root@75a8e243014f:/#

Server Info:
-----
Linux oa222sv-server 6.8.0-62-generic #65-Ubuntu SMP PREEMPT_DYNAMIC Mon May 19 17:15:03 UTC 2025 x86_64 x86_64 GNU/Linux
Filesystem      Size      Used Avail Use% Mounted on
/dev/root        8.7G      2.5G      6.2G 29% /
tmpfs            985M        0      985M   0% /dev/shm
tmpfs            394M      740K      393M   1% /run
tmpfs            5.0M        0       5.0M   0% /run/lock
tmpfs            197M      8.0K      197M   1% /run/user/1000
/dev/sda16       881M     133M     687M  17% /boot
/dev/sda15       105M      6.2M      99M   6% /boot/efi
15:48:56 up 9 days, 8:56,  3 users, load average: 0.01, 0.01, 0.00
             total    used    free   shared  buff/cache   available
Mem:          1.9Gi     434Mi     376Mi       1.1Mi       1.3Gi       1.5Gi
Swap:          0B           0B           0B

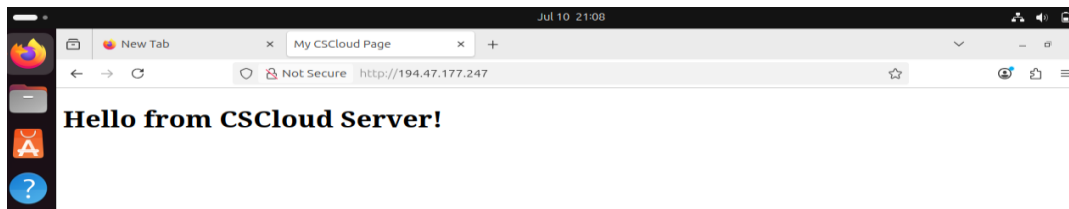
```

Requirements 3

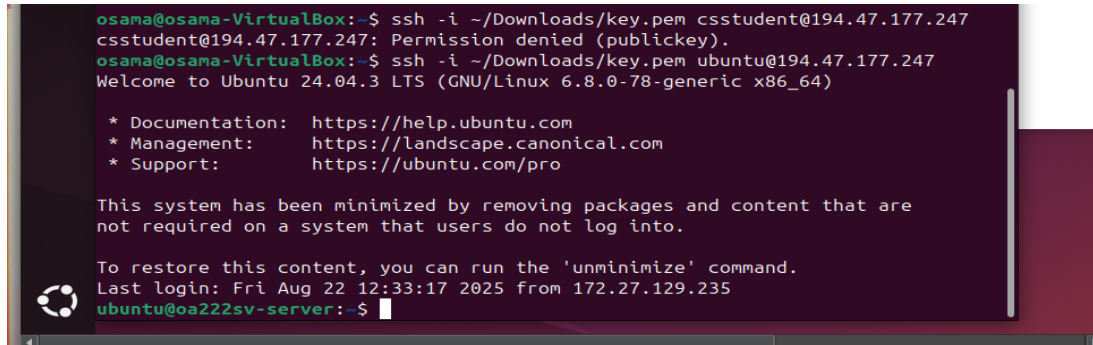
I Chose Ubuntu 24.04 LTS and installed it on VirtualBox.



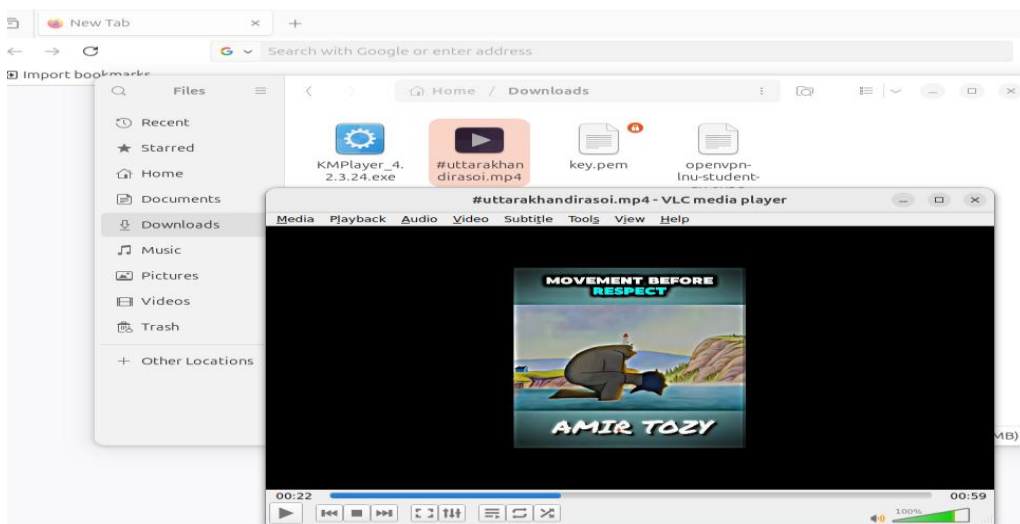
Firefox was installed and tested by browsing and connecting to the CSCloud server.



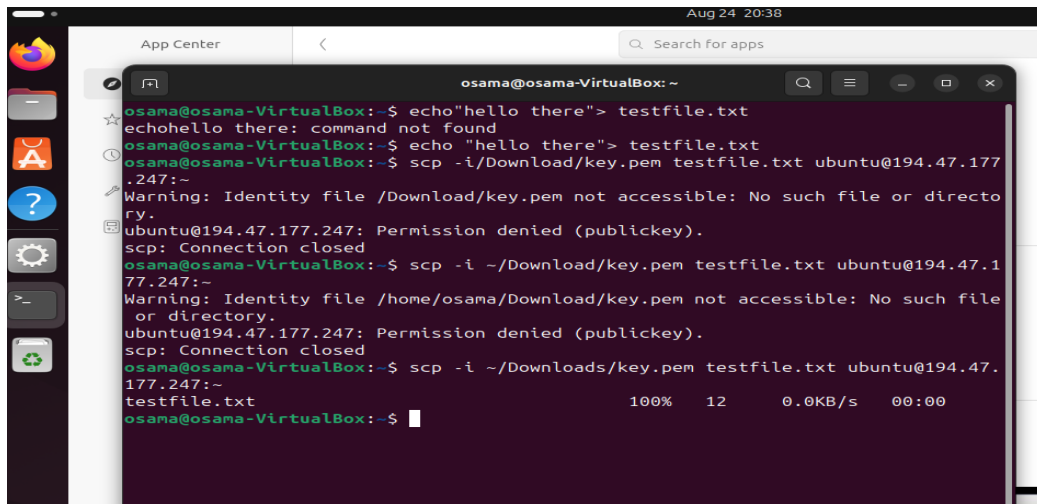
Moved .pem key, corrected file permissions, and connected via SSH:



Installed VLC Media Player via App Center and opened an .mp4 video successfully.

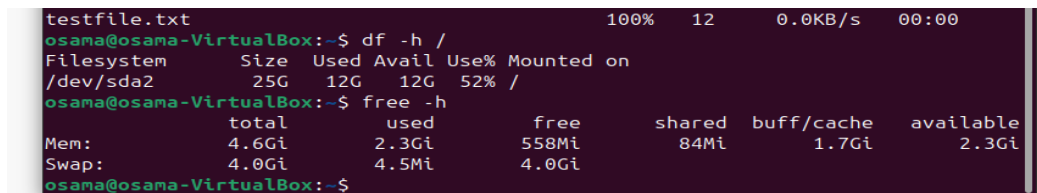


Created testfile.txt and transferred it:

A screenshot of a terminal window titled 'osama@osama-VirtualBox: ~'. The terminal shows the following commands and output:

```
osama@osama-VirtualBox:~$ echo "hello there"> testfile.txt
echohello there: command not found
osama@osama-VirtualBox:~$ echo "hello there"> testfile.txt
osama@osama-VirtualBox:~$ scp -i /Download/key.pem testfile.txt ubuntu@194.47.177.247:~
Warning: Identity file /Download/key.pem not accessible: No such file or directory.
ubuntu@194.47.177.247: Permission denied (publickey).
scp: Connection closed
osama@osama-VirtualBox:~$ scp -i ~/Download/key.pem testfile.txt ubuntu@194.47.177.247:~
Warning: Identity file /home/osama/Download/key.pem not accessible: No such file or directory.
ubuntu@194.47.177.247: Permission denied (publickey).
scp: Connection closed
osama@osama-VirtualBox:~$ scp -i ~/Downloads/key.pem testfile.txt ubuntu@194.47.177.247:~
testfile.txt                                100% 12      0.0KB/s   00:00
osama@osama-VirtualBox:~$
```

Used df -h and free -h to check disk and RAM usage:

A screenshot of a terminal window showing the output of 'df -h /' and 'free -h'.

```
testfile.txt                                100% 12      0.0KB/s   00:00
osama@osama-VirtualBox:~$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2       25G   12G   12G   52% /
osama@osama-VirtualBox:~$ free -h
              total        used         free       shared  buff/cache   available
Mem:           4.6Gi        2.3Gi        558Mi        84Mi        1.7Gi        2.3Gi
Swap:          4.0Gi         4.5Mi        4.0Gi
```

Ubuntu 24.04 worked well. It was fast enough, and I could install apps, play videos, use SSH, and transfer files, it was a good option.

TIL

Today I learned how to connect to a server using SSH and a key file. I also learned how with MariaDB and Apache. It was easy to follow some steps, but some parts like Docker and CGI were a bit confusing, and I had to try a few times to get everything working. I also learned how to write bash scripts and how to run them from a web server.

YouTube Video Presentation

Here is the link to my recorded presentation:

<https://www.youtube.com/watch?v=CsbfWI0xZbc>